



Yakeen Plus (2025)

PRACTICE TEST- 06

DURATION :200 Minutes

DATE : 23/10/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (2)
2. (2)
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CHEMISTRY

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199. (3)
200. (1)

SECTION-(I) PHYSICS

1. (2)
When mass 700 gm is removed, the left out mass (500 + 400) gm oscillates with a period of 3 sec

$$\therefore 3 = t = 2\pi\sqrt{\frac{(500+400)}{k}} \quad \dots(i)$$

When 500 gm mass is also removed, the left out mass is 400 gm.

$$\therefore t' = 2\pi\sqrt{\frac{400}{k}} \quad \dots(ii)$$

$$\Rightarrow \frac{3}{t'} = \sqrt{\frac{900}{400}} \Rightarrow t' = 2 \text{ sec}$$
2. (2)

$$\frac{3V}{2L_1} = \frac{3V}{4L} \Rightarrow L_1 = 2L$$
3. (2)

$$U = \frac{1}{2}Kx^2 \text{ but } T = Kx$$

So, energy stored = $\frac{1}{2} \frac{(Kx)^2}{K} = \frac{1}{2} \frac{T^2}{K}$
4. (1)
With mass m_2 alone, the extension of the spring l is given as $m_2g = kl \quad \dots(i)$

With mass $(m_1 + m_2)$, the extension l' is given by $(m_1 + m_2)g = k(l + \Delta l) \quad \dots(ii)$

The increase in extension is Δl which is the amplitude of vibration. Subtracting (i) from (ii), we get

$$m_1g = k\Delta l \text{ or } \Delta l = \frac{m_1g}{k}$$
5. (2)
With respect to the block the springs are connected in parallel combination.

$$\therefore \text{Combined stiffness } k = k_1 + k_2 \text{ and}$$

$$n = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}}$$
6. (1)

$$n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \Rightarrow \frac{n}{n'} = \sqrt{\frac{k}{m} \times \frac{m'}{K'}} = \sqrt{\frac{k}{m} \times \frac{2m}{2K}} = 1$$

$$\Rightarrow n' = n$$
7. (3)
The effective acceleration in a lift descending with acceleration $\frac{g}{3}$ is $g_{eff} = g - \frac{g}{3} = \frac{2g}{3}$

$$\therefore T = 2\pi\sqrt{\frac{L}{g_{eff}}} = 2\pi\sqrt{\frac{L}{(2g/3)}} = 2\pi\sqrt{\frac{3L}{2g}}$$
8. (3)
Time period is independent of mass of pendulum.
9. (4)
Given max velocity $\omega a = 1$ and maximum acceleration $\omega^2 a = 1.57$

$$\therefore \frac{\omega^2 a}{\omega a} = 1.57 \Rightarrow \omega = 1.57$$

$$\Rightarrow \frac{2\pi}{T} = 1.57 \Rightarrow T = 4 \text{ sec}$$
10. (1)
At mean position, the kinetic energy is maximum.

Hence $\frac{1}{2}ma^2\omega^2 = 16$

On putting the values, we get

$$\omega = 10$$

$$\Rightarrow T = \frac{2\pi}{\omega} = \frac{\pi}{5} \text{ sec}$$
11. (4)
From the given equitation $\omega = 2\pi n = 4\pi \Rightarrow n = 2 \text{ Hz}$
12. (1)

$$x = A \cos\left(\omega t + \frac{\pi}{4}\right) \text{ and}$$

$$v = \frac{dx}{dt} = -A\omega \sin\left(\omega t + \frac{\pi}{4}\right)$$

For maximum speed,

$$\sin\left(\omega t + \frac{\pi}{4}\right) = 1$$

$$\Rightarrow \omega t + \frac{\pi}{4} = \frac{\pi}{2} \text{ or } \omega t = \frac{\pi}{2} - \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4\omega}$$
13. (2)
The sound waves in air are longitudinal while the light waves are transverse.
14. (1)

$$I \propto \frac{1}{r^2}$$

$$I_P = \frac{k}{2^2}$$

$$I_Q = \frac{k}{3^2}$$

$$I_P : I_Q = 9 : 4$$

15. (2)

$$y_1 = 4\sin 500\pi t \text{ and } y_2 = 2\sin 506\pi t$$

$$\omega_1 = 2\pi f_1 = 500\pi$$

$$f_1 = 250 \text{ Hz}$$

$$\text{Similarly, } 2\pi f_2 = 506\pi$$

$$f_2 = 253 \text{ Hz}$$

$$\text{Beats} = f_2 - f_1 = 3 \text{ Hz}$$

$$3 \text{ beats in 1 second then number of beats in 1 minute} = 3 \times 60 = 180$$

16. (2)

$$f' = f \left(\frac{v + v_0}{v - v_s} \right) \quad \because v_0 = v_s$$

$$= 600 \left(\frac{330 + 30}{330 - 30} \right) = 2(360) = 720 \text{ Hz}$$

17. (1)

$$\frac{(n+1)}{n} = \frac{420}{315}$$

$$\frac{n+1}{n} = \frac{84}{63}$$

$$63n + 63 = 84n$$

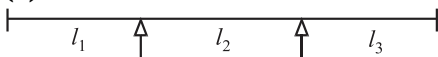
$$63 = 21n$$

$$n = 3$$

$$\frac{1}{3} \text{ of } 315 \text{ is fundamental.}$$

$$f = 105$$

18. (1)



$$l = l_1 + l_2 + l_3$$

$$\frac{1}{2n} \sqrt{\frac{T}{\mu}} = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}} + \frac{1}{2n_2} \sqrt{\frac{T}{\mu}} + \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$

$$\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$$

$$n_1 = \frac{1}{n l_1} \sqrt{\frac{T}{\mu}} \Rightarrow l_1 = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}}$$

$$n_2 = \frac{1}{2l_2} \sqrt{\frac{T}{\mu}} \Rightarrow l_2 = \frac{1}{2n_2} \sqrt{\frac{T}{\mu}}$$

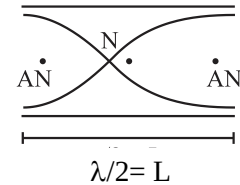
$$n_3 = \frac{1}{2l_3} \sqrt{\frac{T}{\mu}} \Rightarrow l_3 = \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$

$$n_1 = \frac{1}{n l_1} \sqrt{\frac{T}{\mu}} \Rightarrow l_1 = \frac{1}{2n_1} \sqrt{\frac{T}{\mu}}$$

$$n_2 = \frac{1}{2l_2} \sqrt{\frac{T}{\mu}} \Rightarrow l_2 = \frac{1}{2n_2} \sqrt{\frac{T}{\mu}}$$

$$n_3 = \frac{1}{2l_3} \sqrt{\frac{T}{\mu}} \Rightarrow l_3 = \frac{1}{2n_3} \sqrt{\frac{T}{\mu}}$$

19. (3)



$$\lambda = 2L$$

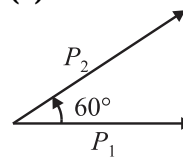
$$f = \frac{v}{\lambda} = \frac{v}{2L}$$

$$f = \frac{nv}{2L}$$

$$N = 1, 2, 3, 4, \dots$$

- Odd and even both harmonic will be present.
- Pressure variation is minimum at antinode.

20. (1)



$$p_{\text{net}} = \sqrt{3}p = \sqrt{3}ql$$

21. (3)

$$\phi_E = \vec{E} \cdot \vec{A} = (5\hat{i} + 8\hat{j} + 9\hat{k}) \cdot 10\hat{i} = 50 \text{ units}$$

22. (4)

$$a = \frac{qE}{m}$$

$$\because E_A > E_B > E_C > E_D$$

Then at point D acceleration of the test charge will be minimum.

23. (3)

$$R = n^{1/3}r = (27)^{1/3}r = 3r$$

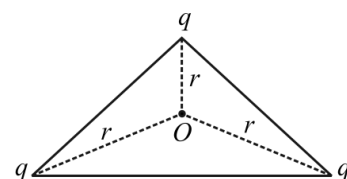
$$\text{Potential of small drop } V = \frac{kq}{r}$$

$$\text{Potential of the big drop } (V') = \frac{kQ}{R} = \frac{k \cdot 27q}{3r}$$

$$V' = 9V$$

$$V' = 9 \times 220 = 1980 \text{ Volt}$$

24. (3)



$$V = k \frac{Q}{r} \times 3 \neq 0$$

$$\text{But } E = 0$$

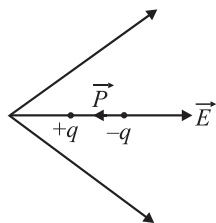
25. (1)

$$\phi = \frac{\Sigma Q}{\epsilon_0}$$

$$\phi = \frac{q - q + q + q - q + q}{\epsilon_0} = \frac{2q}{\epsilon_0}$$

26. (2)

Potential energy of electric dipole in external electric field $U = -\vec{P} \cdot \vec{E}$



Angle between electric field and electric dipole is 180° .

$$U = -PE \cos \theta$$

$$U = -PE \cos 180^\circ$$

$$U = +PE$$

On moving towards right electric field strength decrease therefore potential energy decrease. Net force on electric dipole is towards right and net torque acting on it is zero. So, it will move towards right.

27. (3)

Now, the electrostatic force acting between the charges is given by,

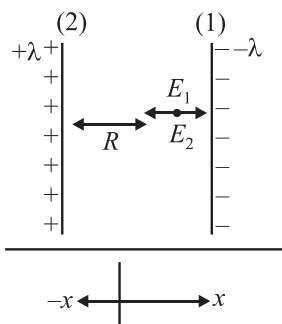
$$F = \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$$

$$= \frac{9 \times 10^9 \times 2 \times 2 \times 10^{-12}}{3^2} = 4 \times 10^{-3} \text{ N}$$

28. (3)

Charge with maximum curved path has highest charge to mass ratio.

29. (3)



Electric field due to line charge (1)

$$\vec{E}_1 = \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

Electric field due to line charge (2)

$$\vec{E}_2 = \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

$$\vec{E}_{\text{net}} = \vec{E}_1 + \vec{E}_2$$

$$= \frac{\lambda}{2\pi\epsilon_0 R} \hat{i} + \frac{\lambda}{2\pi\epsilon_0 R} \hat{i}$$

$$= \frac{\lambda}{\pi\epsilon_0 R} \hat{i} \text{ N/C}$$

30. (1)

$$E = -\frac{\partial V}{\partial x}$$

$$\therefore V = 4x^2$$

$$E = -\frac{\partial}{\partial x} 4x^2$$

$$E = -8x$$

At point (1m, 0, 2m)

$$E = -8 \times 1 = -8$$

$$E = -8 \text{ V/m}$$

31. (2)

Charge density (σ) is same is false but charge same.

$$\text{Charge density, } \sigma = \frac{Q}{A}$$

In case of charge distributed uniformly a conductor has to be equipotential under static condition.

32. (2)

Electric lines of force never cross each other. If they cross each other, then at that point, we get two directions of electric field at that point, which is not possible. So, the assertion is false. The resultant electric field at a point is a vector sum of the electric fields at that point.

33. (3)

When an electric dipole is placed in a uniform electric field at an angle θ with the field, the dipole experiences a torque.

The torque produced by two parallel forces qE acting as couple $= \tau$

$$\tau = qE (2l \sin \theta)$$

In case of non-uniform field, force acting on both the ends of the dipole will not be equal. So, there will be a combination of couple and a net force. In this way, dipole will have both rotational as well as linear motion.

So, both assertion and reason are true. Reason also explains the assertion.

34. (1)

$$\Delta V = - \int E dx = \text{Area under } E - x \text{ graph.}$$

$$\Delta V = (4-2) \times 10 + \frac{1}{2} \times (6-4) \times 10$$

$$= 30 \text{ Volt.}$$

35. (2)

Theory

36. (4)

$$W = \Delta U$$

$$= U_f - U_i$$

$$= QV_f - QV_i$$

$$= \frac{1}{4\pi\epsilon_0} \left(\frac{-Qq}{l} \right) + \frac{1}{4\pi\epsilon_0} \left(\frac{Qq}{3l} \right)$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Qq}{l} \left(\frac{1}{3} - 1 \right) = -\frac{2}{3} \frac{1}{4\pi\epsilon_0} \frac{Qq}{l}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Qq}{l} \left(\frac{1}{3} - 1 \right) = -\frac{1}{6\pi\epsilon_0} \frac{Qq}{l}$$

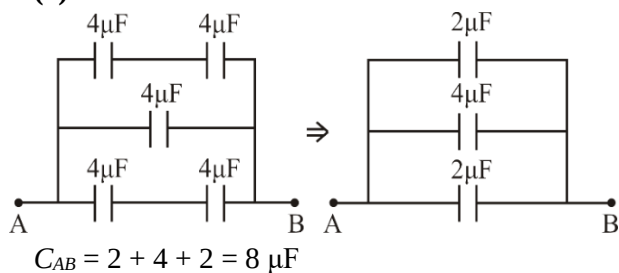
37. (2)

$$C_1 = \frac{K_1 \epsilon_0 A}{d/2} \quad C_2 = \frac{K_2 \epsilon_0 A}{d/2}$$

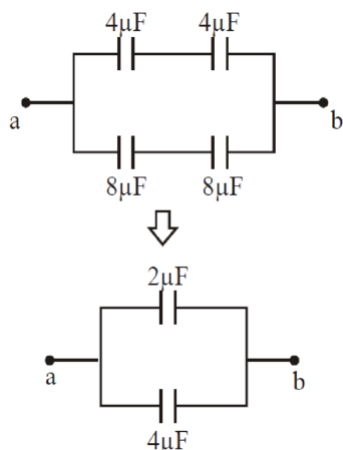
$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow C = \frac{2\epsilon_0 AK_1 K_2}{d(K_1 + K_2)}$$

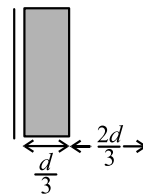
38. (1)



39. (1)



40. (3)

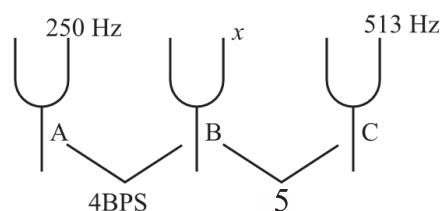


$$C_1 = \frac{KA\epsilon_0}{d/3}, C_2 = \frac{3A\epsilon_0}{2d}$$

$$\therefore C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{6 \times \frac{3}{2} \left(\frac{A\epsilon_0}{d} \right)^2}{\left(6 + \frac{3}{2} \right) \frac{A\epsilon_0}{d}}$$

$$= \frac{18 A\epsilon_0}{15 d} = \frac{6 A\epsilon_0}{5 d}$$

41. (4)



$$BPS = |f_A - f_x|$$

$$f_x = 254 \text{ Hz, } 246 \text{ Hz}$$

$$BPS = |f_C - 2f_x|$$

$$2f_x = 518 \text{ Hz} = 508 \text{ Hz}$$

OR

$$f_x = 259 \text{ Hz} = 254 \text{ Hz}$$

$$f_x = 254 \text{ Hz}$$

42. (4)

$$y = A \sin(kx - \omega t) = 1 \sin\left(\frac{2\pi}{\lambda} x - 2\pi f t\right)$$

$$y = \sin\left(\frac{2\pi}{2\pi} x - 2\pi \times \frac{1}{\pi} t\right)$$

$$y = \sin(x - 2t)$$

43. (2)

$$y = 3 \sin \frac{\pi}{2} (50t - x)$$

$$v_{\max} = A\omega$$

$$\text{and } v = f\lambda$$

$$\frac{v_{\max}}{v} = \frac{A\omega}{f\lambda}$$

$$\frac{v_{\max}}{v} = \frac{A2\pi f}{f\lambda} = \frac{2\pi A}{\lambda}$$

$$\therefore k = \frac{2\pi}{\lambda}$$

$$\frac{v_{\max}}{v} = kA = \frac{\pi}{2} \times 3 = \frac{3\pi}{2}$$

44. (4)

$$I_1 = 9.75 \text{ cm}$$

$$I_2 = 31.25 \text{ cm}$$

$$I_3 = 52.75 \text{ cm}$$

$e =$ end correction

$$\frac{\lambda}{4} + e = 9.75 \text{ cm}$$

$$\frac{3\lambda}{4} + e = 31.25 \text{ cm}$$

$$\frac{3\lambda}{4} - \frac{\lambda}{4} = 31.25 - 9.75$$

$$\frac{\lambda}{2} = 21.5$$

$$\lambda = 43 \text{ cm}$$

$$v = \lambda \times f$$

$$v = 43 \times 800$$

$$v = 34400 \times 10^{-2}$$

$$v = 344 \text{ m/s}$$

45. (2)

$$v = 2(v)[L_2 - L_1]$$

$$= 2 \times 320 [73 - 20] \times 10^{-2}$$

$$= 339.2 \text{ m s}^{-1}$$

$$= 339 \text{ m/s}$$

46. (2)

Two successive frequencies of closed pipe

$$\frac{nv}{4l} = 220 \quad \dots(i)$$

$$\frac{(n+2)v}{4l} = 260 \quad \dots(ii)$$

Dividing (ii) by (i), we get

$$\frac{n+2}{n} = \frac{260}{220}$$

$$= \frac{13}{11}$$

$$11n + 22 = 13n$$

$$n = 11$$

$$\text{So, } 11 \frac{v}{4l} = 220$$

$$\frac{v}{4l} = 20$$

So fundamental frequency is 20 Hz.

47. (4)

$(n-1)$ and $(n+1)$ suppose to form frequency n

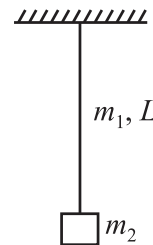
n and n will be at resonance

$n-1$ and $n \rightarrow$ produce 1 beat

$n+1$ and $n \rightarrow$ produce 1 beat

Number of beats formed are '2'.

48. (3)



$$\lambda = \frac{v}{f} \left(v = \sqrt{\frac{T}{\mu}} \right)$$

$$\frac{\lambda_2}{\lambda_1} = \frac{v_2}{v_1}$$

$$= \sqrt{\frac{T_2}{T_1}}$$

$$= \sqrt{\frac{(m_1 + m_2)}{m_2}}$$

49. (4)

$$\eta_c = \frac{v}{4I_c}$$

$$\eta_0 = 3 \left(\frac{v}{2I_0} \right) = 2^{\text{nd}} \text{ overtone}$$

$$\frac{2V}{2I_0} = \frac{v}{4I_c}$$

$$\Rightarrow 6I_c = I_0$$

$$6 \times 20 = I_0$$

$$120 = I_0$$

50. (3)

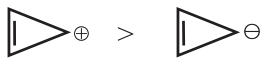
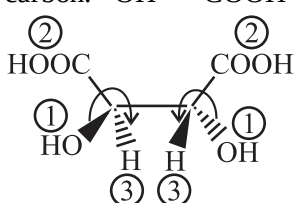
$$\text{If } y = f(ax \pm bt)$$

$$\text{then } v = \left| \frac{b}{a} \right|$$

$$y = \frac{9}{3 + (x - 10t)^2}$$

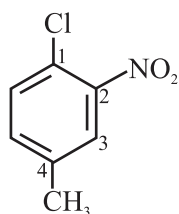
$$v = 10 \text{ m/s}$$

SECTION-(II) CHEMISTRY

51. (3)
 $\text{CH}_3-\text{CH}=\underset{\text{OH}}{\text{C}}-\text{CH}_2-\text{NH}_2$
 1-Aminobut-2-en-2-ol
 [New NCERT Class 11th Page No. 267]
52. (4)

 (Aromatic) (Antiaromatic)
 [New NCERT Class 11th Page No. 321]
53. (3)
 n-pentane and neopentane are chain isomers to each other.
 [New NCERT Class 11th Page No. 297]
54. (1)
 Positive charge on an atom makes them electron deficient as compared to neutral atom which results in withdrawal of electron from nearby group,
 Hence, $-I$ effect increases
 [New NCERT Class 11th Page No. 274]
55. (2)
 2-Methylpentane-2, 4-diol
 [New NCERT Class 11th Page No. 267]
56. (2)
 Hyperconjugation is σ - π conjugation also known as no bond resonance.
 [New NCERT Class 11th Page No. 278]
57. (3)
 $sp > sp^2 > sp^3$ ($-I$ effect order)
 (50%) (33%) (25%) (s-character)
 $s\text{-character} \propto \text{electronegativity of carbocation}$
 [New NCERT Class 11th Page No. 274]
58. (2)
 (1) There is no stereo center.
 (2) There is one stereo center. All atoms attached to carbon atom are different as you can see in the diagram. So, it is optically active.
 (3) There is no stereo center as there are two same (CH_3) groups on carbon.
 (4) There is no stereo center.
 [New NCERT Class 11th Page No. 270]
59. (4)
 sp, sp^3, sp^2 and sp^3
 [New NCERT Class 11th Page No. 257]
60. (2)
 • Nucleophiles are electron rich species and donate their electrons to the electron deficient species.
 • Lewis base is a base which donates a pair of electrons.
 [New NCERT Class 11th Page No. 272]
61. (4)
 1-Chloropropane and 2-Chloropropane are position isomers of each other.
 $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$ and $\text{CH}_3-\overset{\text{Cl}}{\underset{|}{\text{CH}}}-\text{CH}_3$
 [New NCERT Class 11th Page No. 270]
62. (1)
 The correct bond-line formula of 3-Ethyl-4-methylpent-1-ene is given in option (1).
 [New NCERT Class 11th Page No. 259]
63. (1)
 Benzylic carbocation is more stable than allylic carbocation.
 [New NCERT Class 11th Page No. 276]
64. (2)
 $\begin{array}{c} \text{CH}_2-\text{CO}_2\text{H} \\ | \\ \text{*CH}-\text{CO}_2\text{H} \\ | \\ \text{HO}-\text{*CH}-\text{CO}_2\text{H} \end{array}$
 Number of stereoisomers = $2^n = 2^2 = 4$
 [New NCERT Class 11th Page No. 270]
65. (1)
 Priority order of groups attached with chiral carbon: $-\text{OH} > -\text{COOH} > -\text{CH}(\text{OH})\text{COOH} > -\text{H}$

 [New NCERT Class 11th Page No. 270]
66. (3)
 Meso compound possess either plane or centre of symmetry. They do not possess both plane of symmetry and centre of symmetry and also they are internally compensated form.
 [New NCERT Class 11th Page No. 270]

67. (2)

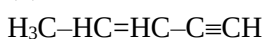
The IUPAC name of given compound is



1-Chloro-4-methyl-2-nitrobenzene

[New NCERT Class 12th Page No. 269]

68. (1)



10 σ bonds and 3 π bonds

[New NCERT Class 11th Page No. 257]

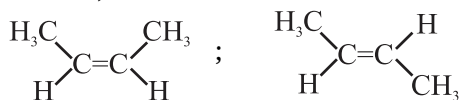
69. (2)

Total Carbon atoms in isomer should be 8 as in given compound total Carbon atoms is 8. But in option (2), Carbon atoms is 7. So, it can not be isomer of given compound.

[New NCERT Class 11th Page No. 270]

70. (2)

$\text{C}_2\text{H}_5\text{CHO}$; CH_3COCH_3 (A pair of functional isomers)



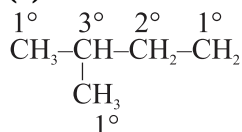
(A pair of geometrical isomers)

$\text{CH}_3\text{OC}_3\text{H}_7$; $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ (A pair of metamers)

$\text{H}_2\text{C}=\text{CHOH}$: CH_3CHO (A pair of tautomers)

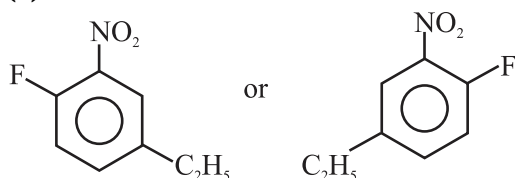
[New NCERT Class 11th Page No. 270-271]

71. (3)



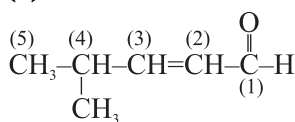
[New NCERT Class 11th Page No. 297]

72. (4)



[New NCERT Class 11th Page No. 269-270]

73. (4)



4-Methylpent-2-enal

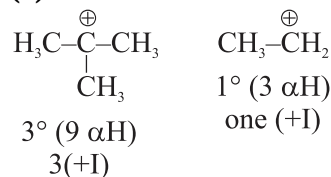
[New NCERT Class 11th Page No. 267]

74. (1)

The formula of organic compound shown with single, double and triple bonds between each atom is known as complete formula.

[New NCERT Class 11th Page No. 259]

75. (1)

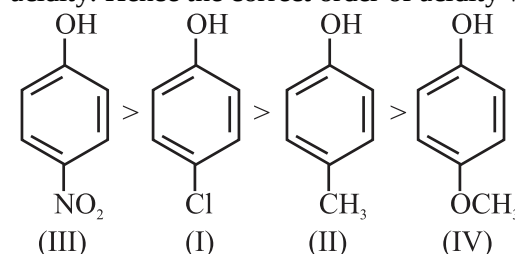


$\therefore 3^\circ$ Carbocation > 1° Carbocation (stability)

[New NCERT Class 11th Page No. 274]

76. (3)

Electron withdrawing substituents like $-\text{NO}_2$, $-\text{Cl}$ increase the acidity of phenol while electron releasing substituents like $-\text{CH}_3$, $-\text{OCH}_3$ decrease acidity. Hence the correct order of acidity will be:

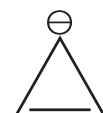


[New NCERT Class 11th Page No. 276]

77. (1)



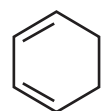
is aromatic ion.



is antiaromatic ion.



is antiaromatic ion.



Nonaromatic.

Hence (1) being aromatic is most stable.

[New NCERT Class 11th Page No. 321]

78. (1)

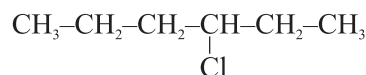
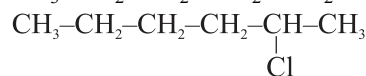
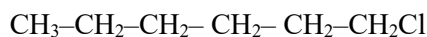


is most acidic because it generates most stable aromatic conjugate base.

[New NCERT Class 11th Page No. 321]

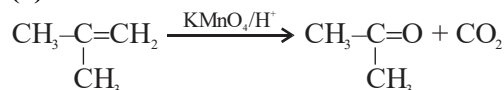
79. (2)

In n-hexane, there is a symmetry of the structure at third and fourth carbon. So, chlorine can be attached to first, second and third carbon to give 3 monochloro derivatives. These monochloro derivatives are:



[New NCERT Class 11th Page No. 303]

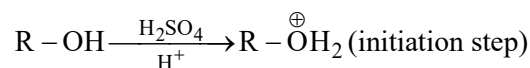
80. (1)



[New NCERT Class 11th Page No. 313]

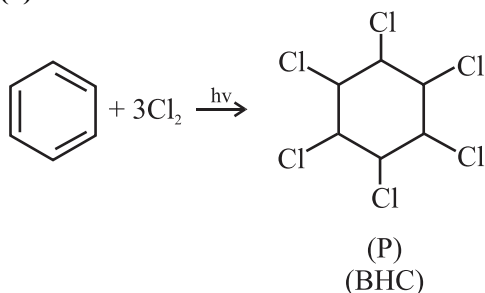
81. (1)

Protonation of alcohol molecule



[New NCERT Class 11th Page No. 310]

82. (1)



[New NCERT Class 11th Page No. 324]

83. (1)

The reaction of 1,1,2,2 tetrabromopropane with zinc powder in alcohol is a dehalogenation reaction.

Zinc (Zn) removes all the bromine atoms and alkyne is formed.

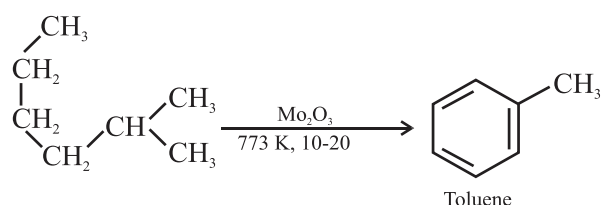
The concerned reaction is as follows;



The product formed is propyne.

[New NCERT Class 11th Page No. 316]

84. (2)



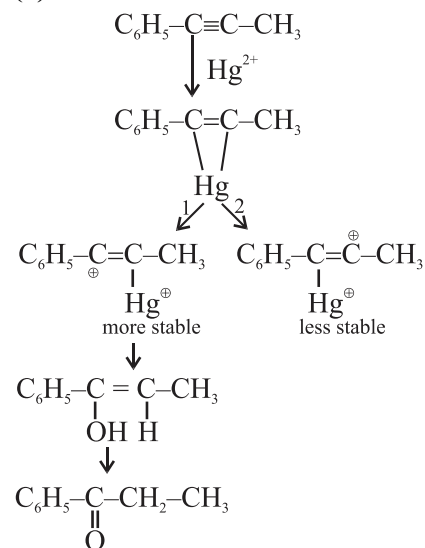
[New NCERT Class 11th Page No. 304]

85. (4)

More the number of alpha hydrogens, more will be the hyperconjugation structures, more will be the stability, and hence lesser heat of hydrogenation will be evolved.

[New NCERT Class 11th Page No. 277]

86. (1)



[New NCERT Class 11th Page No. 317]

87. (3)

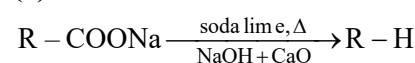
Statement I: The melting point of neo-pentane is greater than that of n-pentane but the boiling point of n-pentane is more than that of neo-pentane.

Statement II: Melting point depends upon packing in crystal lattice whereas boiling point depends upon the surface area of the molecule.

Both Statement I and Statement II are correct.

[New NCERT Class 11th Page No. 302]

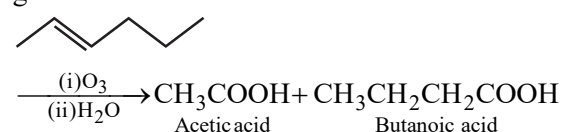
88. (1)



[New NCERT Class 11th Page No. 301]

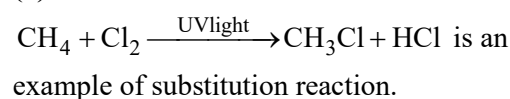
89. (1)

2-hexene on ozonolysis in the presence of water give acetic acid and butanoic acid.



[New NCERT Class 11th Page No. 313]

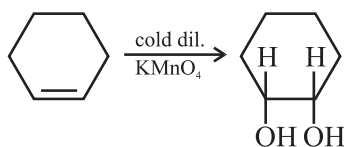
90. (2)



[New NCERT Class 11th Page No. 303]

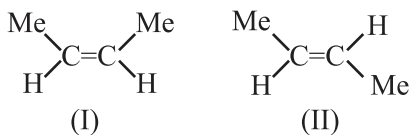
91. (3)

Cold alkaline KMnO_4 converts alkene into cis-diol.



[New NCERT Class 11th Page No. 313]

92. (4)



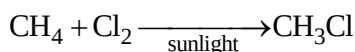
- (1) Dipole moment $\text{I} > \text{II}$
- (2) Boiling point $\text{I} > \text{II}$
- (3) Solubility in H_2O $\text{I} > \text{II}$
- (4) Melting point $\text{II} > \text{I}$

[New NCERT Class 11th Page No. 309]

93. (3)

Decarboxylation $\Rightarrow \text{RCOONa} + \text{sodalime} \rightarrow$

wurtz reaction $\Rightarrow \text{Alkyl halide} + \text{Na} \rightarrow$



$\Rightarrow$ Free radical substitution

Saturated hydrocarbons $\Rightarrow$ Paraffins

[New NCERT Class 11th Page No. 301]

94. (4)



(Cyclopenta-1, 3-diene)

does not follow Huckel's rule so it is non aromatic in nature.

[New NCERT Class 11th Page No. 321]

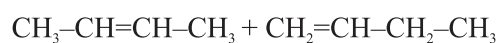
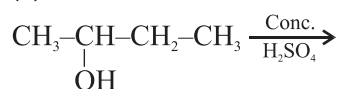
95. (2)



(Antiaromatic)

[New NCERT Class 11th Page No. 321]

96. (3)

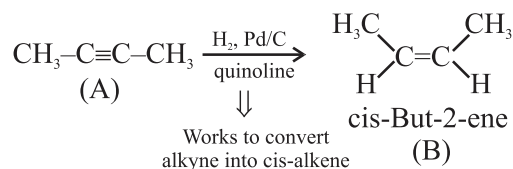


Major

Minor

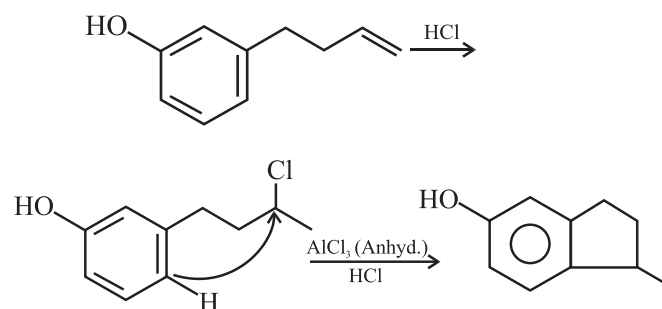
[New NCERT Class 11th Page No. 310]

97. (2)



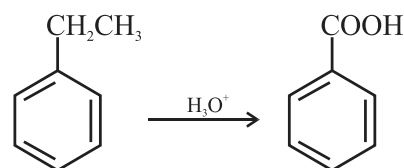
[New NCERT Class 11th Page No. 309]

98. (4)



[New NCERT Class 11th Page No. 310]

99. (3)



Alkaline KMnO_4 is used to oxidise

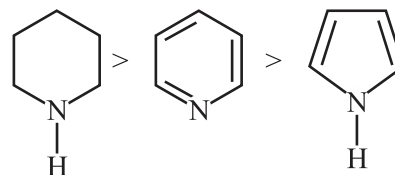
$-\text{CH}_2-\text{CH}_3$ group into $-\text{COOH}$ group

[New NCERT Class 11th Page No. 304]

100. (2)

Basic strength increases as the electron donating capacity of nitrogen increases.

In compound III, the nitrogen has one lone pairs of electrons for donation whereas, in compound II, the lone pair of electrons are delocalized in the aromatic ring.



[New NCERT Class 11th Page No. 276]

SECTION-(III) BOTANY

101. (3)

Xylem parenchyma cells are living and thin-walled, and their cell walls are made up of cellulose. They store food materials in the form of starch or fat, and other substances like tannins.

[Old NCERT Class 11th Page No. 87]

102. (3)

Stomata regulate the process of transpiration and gaseous exchange.

[New NCERT Class 11th Page No. 72, 77]

103. (2)

Pericycle is present on the inner side of the endodermis and above the phloem in the form of semi-lunar patches of sclerenchyma.

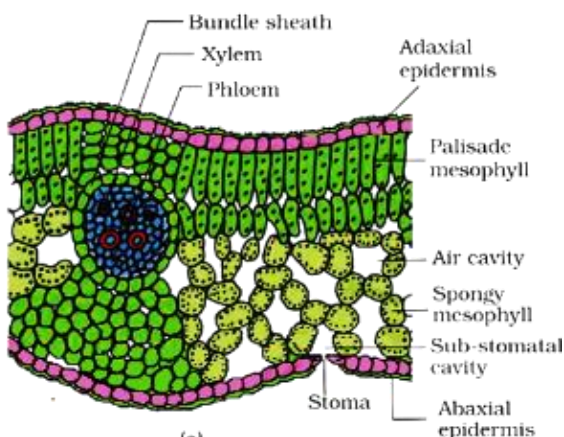
[New NCERT Class 11th Page No. 75, 76]

104. (3)

Different organs in a plant show differences in their internal structure. Internal structures show adaptations to diverse environments.

[New NCERT Class 11th Page No. 71]

105. (2)



A– Epidermis, B– Palisade mesophyll

[New NCERT Class 11th Page No. 76]

106. (4)

The innermost layer of the cortex is called endodermis. It comprises a single layer of barrel shaped cells without any intercellular spaces.

[New NCERT Class 11th Page No. 74]

107. (1)

Sclerenchyma consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits. They are usually dead and without protoplasts.

[Old NCERT Class 11th Page No. 86]

108. (2)

Ground tissue consists of simple tissues such as parenchyma, collenchyma and sclerenchyma. Parenchymatous cells are usually present in cortex, pericycle, pith and medullary rays, in the primary stems and roots.

[New NCERT Class 11th Page No. 72]

109. (1)

When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, the arrangement is called radial such as in roots.

[New NCERT Class 11th Page No. 73]

110. (3)

The epidermis is the outermost layer of the primary plant body. It is made up of elongated, compactly arranged cells, which form a continuous layer. Epidermis is usually single layered. Epidermal cells are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole.

[New NCERT Class 11th Page No. 71, 72]

111. (1)

The parenchymatous cells in dicotyledonous root which lie between the xylem and the phloem are called conjunctive tissue.

[New NCERT Class 11th Page No. 74]

112. (3)

Cuticle	Waxy layer
Trichomes	Multicellular
Root hair	Unicellular
Stomata	Pore

[New NCERT Class 11th Page No. 72]

113. (1)

The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material suberin in the form of casparian strips.

[New NCERT Class 11th Page No. 74]

114. (4)

The parallel venation, bulliform cells are present in monocotyledonous (isobilateral) leaves.

[New NCERT Class 11th Page No. 76]

115. (1)

The monocotyledonous stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath, and a large, conspicuous parenchymatous ground tissue.

[New NCERT Class 11th Page No. 76, 77]

116. (1)

The vascular bundles are surrounded by a layer of thick walled bundle sheath cells.

[New NCERT Class 11th Page No. 77]

117. (1)

A large number of rounded, parenchymatous cells with large intercellular spaces which occupy the central portion of the stem constitute the pith.

[New NCERT Class 11th Page No. 76]

118. (3)

Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate.

The phosphorylated form of glucose isomerises to produce fructose-6-phosphate.

[New NCERT Class 11th Page No. 155]

119. (2)

The compounds that are oxidised during this process are known as respiratory substrates.

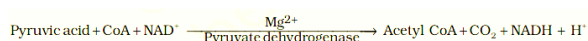
[New NCERT Class 11th Page No. 154]

120. (1)

In tricarboxylic acid cycle, during the conversion of succinyl-CoA to succinic acid a molecule of GTP is synthesised. This is a substrate level phosphorylation. In a coupled reaction GTP is converted to GDP with the simultaneous synthesis of ATP from ADP.

[New NCERT Class 11th Page No. 156, 159]

121. (2)



[New NCERT Class 11th Page No. 158]

122. (1)

Cytochrome *c* is a small protein attached to the outer surface of the inner membrane and acts as a mobile carrier for transfer of electrons between complex III and IV.

[New NCERT Class 11th Page No. 160]

123. (3)

Complex I	NADH dehydrogenase
Complex II	Succinate dehydrogenase
Complex III	Cytochrome <i>bc</i> ₁ complex
Complex IV	Cytochrome <i>c</i> oxidase

[New NCERT Class 11th Page No. 160]

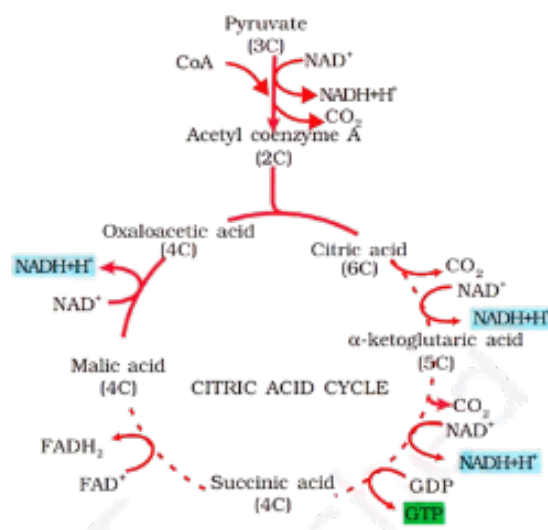
124. (4)

Fermentation accounts for only a partial breakdown of glucose whereas in aerobic respiration it is completely degraded to CO₂ and H₂O.

[New NCERT Class 11th Page No. 162]

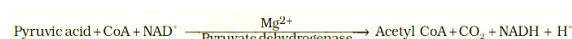
125. (4)

The connecting link between EMP pathway (glycolysis) and citric acid cycle is acetyl CoA. It links the end product of glycolysis (pyruvate) to the entry point of the Krebs's cycle.



[New NCERT Class 11th Page No. 158, 159]

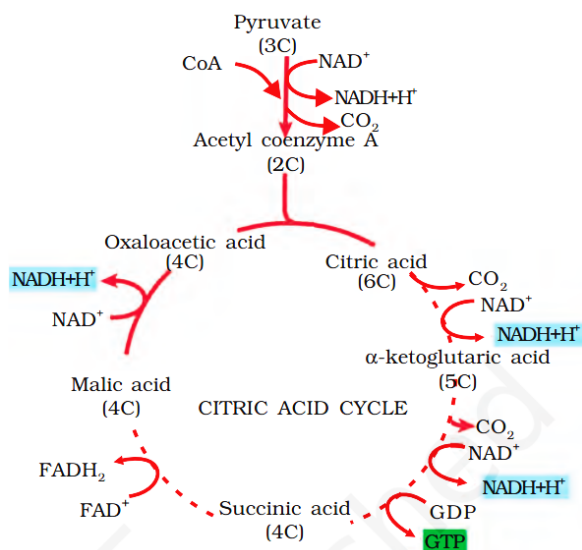
126. (2)



[New NCERT Class 11th Page No. 158]

127. (4)

The crucial events in aerobic respiration are: • The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 .



[New NCERT Class 11th Page No. 159]

128. (4)

Acetyl CoA	2-Carbon compound
Fructose	6-Carbon compound
Pyruvic acid	3-Carbon compound
Oxaloacetic acid	4-Carbon compound

[New NCERT Class 11th Page No. 156, 159]

129. (3)

For aerobic respiration to take place within the mitochondria, the final product of glycolysis, pyruvate is transported from the cytoplasm into the mitochondria. The enzymes of TCA cycle are thus found in mitochondrial matrix and inner mitochondrial membrane.

[New NCERT Class 11th Page No. 158]

130. (1)

Glucose and fructose are phosphorylated to give rise to glucose-6-phosphate by the activity of the enzyme hexokinase. Invertase helps in breakdown of glucose to fructose.

[New NCERT Class 11th Page No. 156]

131. (4)

The Krebs's cycle starts with the condensation of acetyl group with oxaloacetic acid (OAA) and water to yield citric acid.

[New NCERT Class 11th Page No. 158]

132. (3)

In glycolysis, a chain of ten reactions, under the control of different enzymes, takes place to produce pyruvate from glucose.

[New NCERT Class 11th Page No. 156]

133. (4)

Succinate dehydrogenase	Provides reducing equivalents to ubiquinone
Cytochrome <i>c</i> oxidase complex	Contains two coppers centres
Cytochrome <i>c</i>	Attached to the outer surface of inner mitochondrial membrane
Complex V	ATP synthase

[New NCERT Class 11th Page No. 160]

134. (4)

- Trichome - Epidermal tissue system
- Guard cells - Epidermal tissue system
- Xylem - Conducting tissue system
- Root hair - Epidermal tissue system

[New NCERT Class 11th Page No. 72, 73]

135. (1)

In fermentation, say by yeast, the incomplete oxidation of glucose is achieved under anaerobic conditions by sets of reactions where pyruvic acid is converted to CO_2 and ethanol. The enzymes, pyruvic acid decarboxylase and alcohol dehydrogenase catalyse these reactions.

[New NCERT Class 11th Page No. 157]

136. (3)

Subsidiary cells are present in monocotyledonous leaf.

[New NCERT Class 11th Page No. 72]

137. (1)

The monocotyledonous stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath, and a large, conspicuous parenchymatous ground tissue. Peripheral vascular bundles are generally smaller than the centrally located ones.

[New NCERT Class 11th Page No. 76]

138. (3)

Pure proteins are never used as respiratory substrate.

Yeast(fermentation)- Ethanol and CO₂.

[New NCERT Class 11th Page No. 157, 164]

139. (2)

In dicotyledonous stem, the outer hypodermis, consists of a few layers of collenchymatous cells just below the epidermis, which provide mechanical strength to the young stem.

[New NCERT Class 11th Page No. 75]

140. (4)

The monocotyledonous stem has a sclerenchymatous hypodermis, a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath, and a large, conspicuous parenchymatous ground tissue vascular bundles are conjoint and closed. Peripheral vascular bundles are generally smaller than the centrally located ones.

[New NCERT Class 11th Page No. 76]

141. (1)

ATP is synthesised during the conversion of phosphoenolpyruvate to pyruvic acid.

[New NCERT Class 11th Page No. 156]

142. (4)

Pith is parenchymatous with very large intercellular spaces in dicotyledonous stem.

[New NCERT Class 11th Page No. 76]

143. (1)

The monocotyledonous stem has a sclerenchymatous hypodermis.

[New NCERT Class 11th Page No. 76]

144. (4)

On the stem the epidermal hairs are called trichomes. The trichomes in the shoot system are usually multicellular. They may be branched or

unbranched and soft or stiff. They may even be secretory. The trichomes help in preventing water loss due to transpiration.

[New NCERT Class 11th Page No. 72]

145. (1)

The TCA cycle requires the continued replenishment of oxaloacetic acid, the first member of the cycle.

[New NCERT Class 11th Page No. 159]

146. (2)

ATP synthesis	1,3-bisphosphoglycerate to 3-phosphoglyceric acid
Reduction	NAD ⁺ to NADH + H ⁺
Decarboxylation	Release of CO ₂ during alcohol fermentation
Isomerisation	Citrate to isocitrate

[New NCERT Class 11th Page No. 156,158,159]

147. (4)

When xylem and phloem within a vascular bundle are arranged in an alternate manner along the different radii, the arrangement is called radial such as in roots.

[New NCERT Class 11th Page No. 73]

148. (3)

The main purpose of electron transport chain is cycle NADH + H⁺ back to NAD⁺.

[New NCERT Class 11th Page No. 159]

149. (4)

There can be a net gain of 38 ATP molecules during aerobic respiration of one molecule of fructose/glucose.

[New NCERT Class 11th Page No. 161]

150. (1)

All tissues on the inner side of the endodermis such as pericycle, vascular bundles and pith constitute the stele.

[New NCERT Class 11th Page No. 74]

SECTION-(IV) ZOOLOGY**151. (4)**

- The neural system provides a point-to-point rapid coordination among organs.
- The neural coordination is fast but short-lived. As the nerve fibres do not innervate all cells of the body and the cellular functions need to be continuously regulated.
- A special kind of coordination and integration has to be provided. This function is carried out by hormones.

[New NCERT Class 11th Page No. 230, 239]**152. (4)**

The neural organisation is very simple in lower invertebrates. For example, in *Hydra* it is composed of a network of neurons.

[New NCERT Class 11th Page No. 231]**153. (4)**

- Pineal secretes a hormone called melatonin.
- Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body. For example, it helps in maintaining the normal rhythms of sleep-wake cycle, body temperature.
- In addition, melatonin also influences metabolism, pigmentation, the menstrual cycle as well as our defense capability.

[New NCERT Class 11th Page No. 242]**154. (1)**

- The thymus secretes the peptide hormones called thymosins.
- Thymosins play a major role in the differentiation of T-lymphocytes, which provide cell-mediated immunity.

[New NCERT Class 11th Page No. 243]**155. (3)**

The hypothalamus is located in the forebrain, below the thalamus, and plays a critical role in maintaining homeostasis.

[New NCERT Class 11th Page No. 236]**156. (2)**

The axons transmit nerve impulses away from the cell body.

[New NCERT Class 11th Page No. 232]**157. (4)**

Each thyroid follicle is composed of follicular cells, enclosing a cavity. These follicular cells synthesise two hormones, tetraiodothyronine or thyroxine (T₄) and triiodothyronine (T₃).

[New NCERT Class 11th Page No. 242, 243]**158. (2)**

The juxtaglomerular cells of kidney produce a peptide hormone called erythropoietin which stimulates erythropoiesis (formation of RBCs).

[New NCERT Class 11th Page No. 247]**159. (4)**

- Myelinated nerve fibres are found in spinal and cranial nerves.
- Unmyelinated nerve fibre is commonly found in autonomous and the somatic neural systems.

[New NCERT Class 11th Page No. 232]**160. (4)**

The released neurotransmitters bind to their specific receptors, present on the postsynaptic membrane.

[New NCERT Class 11th Page No. 235]**161. (3)**

- Estrogen directly affect transcription by acting on gene.
- Hormones which interact with intracellular receptors (e.g., steroid hormones, iodothyronines, etc.) mostly regulate gene expression or chromosome function by the interaction of the hormone-receptor complex with the genome.

[New NCERT Class 11th Page No. 248]**162. (3)**

Pia mater is in contact with the brain tissue.

[New NCERT Class 11th Page No. 235]**163. (3)**

- The cerebral cortex is referred to as grey matter due to its greyish appearance.
- Resting axonal membrane is comparatively more permeable to potassium ions.

[New NCERT Class 11th Page No. 232, 236]**164. (1)**

Unmyelinated nerve fibre is enclosed by a Schwann cell that does not form a myelin sheath around the axon, and is commonly found in autonomous and the somatic neural systems.

[New NCERT Class 11th Page No. 232]

165. (4)

Cretinism	Thyroid hormone
Addison's disease	Adrenal cortex hormone
Acromegaly	Growth hormone
Diabetes Insipidus	ADH

[New NCERT Class 11th Page No. 241, 242, 243, 244]

166. (3)

- Insulin acts mainly on hepatocytes and adipocytes (cells of adipose tissue) and enhances cellular glucose uptake and utilisation.
- Insulin also stimulates the conversion of glucose to glycogen (glycogenesis) in the target cells.

[New NCERT Class 11th Page No. 245, 246]

167. (3)

The sympathetic nervous system is part of the autonomic nervous system and prepares the body for "fight or flight" responses during stress.

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168. (1)

Somatic neural system relays impulses from the CNS to skeletal muscles while the autonomic neural system transmits impulses from the CNS to the involuntary organs and smooth muscles of the body.

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169. (2)

Atrial wall of heart	ANF
Thyroid gland	T ₃
CCK	Stimulates secretion of pancreatic enzyme and bile
GIP	Inhibits gastric secretion

[New NCERT Class 11th Page No. 242, 247]

170. (3)

- Androgens play a major stimulatory role in the process of spermatogenesis (formation of spermatozoa).
- Androgens act on the central neural system and influence the male sexual behavior (libido).
- These hormones produce anabolic (synthetic) effects on protein and carbohydrate metabolism.

[New NCERT Class 11th Page No. 246]

171. (4)

- The pituitary gland is located in a bony cavity called sella tursica.
- It is divided anatomically into adenohypophysis and neurohypophysis.
- Pars distalis produces adrenocorticotrophic hormone.
- Pars intermedia secretes only one hormone called melanocyte stimulating hormone (MSH).
- The posterior pituitary is under direct neural regulation of the hypothalamus.

[New NCERT Class 11th Page No. 241]

172. (1)

The dorsal portion of the midbrain consists mainly of four round swellings (lobes) called corpora quadrigemina.

[New NCERT Class 11th Page No. 236]

173. (3)

- The nerve fibres of the PNS are of two types.
- Afferent fibers send impulses from tissues to the CNS, while efferent fibers relay signals from the CNS to tissues.

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174. (3)

Zona fasciculata	Middle layer of adrenal cortex
Glucocorticoids	Stimulate lipolysis, RBC production
T-lymphocytes	Cell-mediated immunity
PTH	Increases the Ca ⁺⁺ levels in blood

[New NCERT Class 11th Page No. 243, 245]

175. (2)

Reabsorption of water and electrolytes by distal tubules of kidney and reducing the loss of water through urine (diuresis) done by vasopressin.

[New NCERT Class 11th Page No. 242]

176. (1)

Non-endocrine tissues secrete several hormones called growth factors.

[New NCERT Class 11th Page No. 247]

177. (3)

- Cyclic AMP is a second messenger.
- Estrogen is synthesised and secreted mainly by the growing ovarian follicles.

[New NCERT Class 11th Page No. 246, 248]

178. (4)

Visceral nervous system is the part of the peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia, and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system.

[New NCERT Class 11th Page No. 231]

179. (1)

The ionic gradient across the resting membrane are maintained by the active transport of ions by the sodium-potassium pump which transports 3 Na⁺ outwards for 2 K⁺ into the cell.

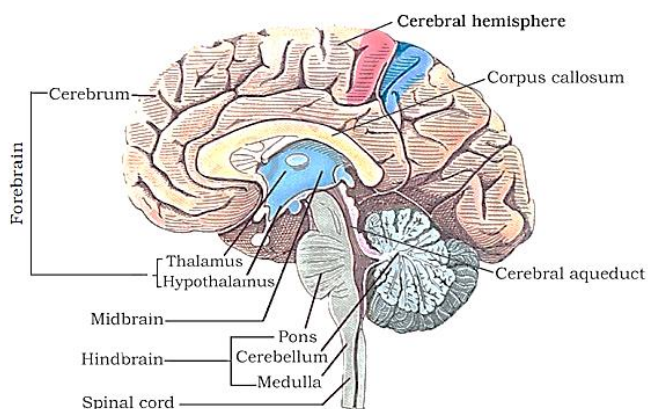
[New NCERT Class 11th Page No. 232]

180. (3)

Prolactin regulates the growth of the mammary glands and formation of milk in them.

[New NCERT Class 11th Page No. 241]

181. (2)



Corpus callosum connects two cerebral hemispheres.

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182. (1)

- The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called the limbic lobe or limbic system.
- Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions (e.g., excitement, pleasure, rage and fear), and motivation.

[New NCERT Class 11th Page No. 236]

183. (2)

- Pars intermedia secretes only one hormone called melanocyte stimulating hormone (MSH).
- FSH, PRL and GH are released by pars distalis region of pituitary.

[New NCERT Class 11th Page No. 241]

184. (4)

In females, LH induces ovulation of fully mature follicles (graafian follicles) and maintains the corpus luteum, formed from the remnants of the graafian follicles after ovulation.

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185. (3)

Pons	Connects different regions of the brain
Hypothalamus	Neuro-secretory cells
Medulla	Controls respiration and gastric secretions
Cerebellum	Provides additional space for neurons

[New NCERT Class 11th Page No. 236]

186. (1)

Three major regions make up the brain stem; mid brain, pons and medulla oblongata. Brain stem forms the connections between the brain and spinal cord.

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187. (4)

Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

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188. (4)

- At rest, the inner side of the membrane is negatively charged (resting potential).
- During an action potential, the inner membrane becomes positively charged (depolarization).
- After the nerve impulse passes, the inner membrane returns to a negative charge (repolarization), restoring the resting potential.

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189. (3)

- Oxytocin acts on the smooth muscles of our body and stimulates their contraction.
- In females, it stimulates a vigorous contraction of uterus at the time of child birth and milk ejection from the mammary gland.

[New NCERT Class 11th Page No. 242]

190. (2)

The cerebrum wraps around a structure called thalamus, which is a major coordinating centre for sensory and motor signaling.

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191. (4)

Medulla	Gastric secretions
Association areas	Memory
Limbic system	Olfaction
Grey matter	Concentration of cell bodies

[New NCERT Class 11th Page No. 236]

192. (3)

The thyroid gland is composed of two lobes which are located on either side of the trachea. Both lobes are interconnected with a thin flap of connective tissue called isthmus.

[New NCERT Class 11th Page No. 242]

193. (4)

Autonomic neural system transmits impulses from the CNS to the involuntary organs and smooth muscles of the body.

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194. (3)

- Oxytocin causes milk ejection from the mammary gland.
- Hormones released during stress are called catecholamines.
- Catecholamines increase the breakdown of glycogen and lipids.

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195. (4)

When a stimulus is applied at any site on the polarised membrane, the membrane becomes freely permeable to Na^+ . This leads to a rapid influx of Na^+ .

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196. (2)

Somatostatin, vasopressin and oxytocin are produced by hypothalamus.

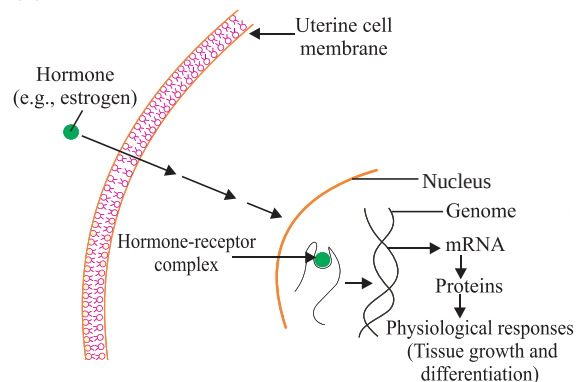
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197. (2)

- The myelinated nerve fibres are enveloped with Schwann cells, which form a myelin sheath around the axon.
- Gaps between the adjacent myelin sheath are called node of Ranvier.

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198. (3)



[New NCERT Class 11th Page No. 249]

199. (3)

- Aldosterone acts mainly at the renal tubules and stimulates the reabsorption of sodium ions and water and excretion of potassium ions and phosphate ions.
- Insulin deficiency leads to decreased glucose uptake by cells.

[New NCERT Class 11th Page No. 243, 245]

200. (1)

Neurohypophysis (pars nervosa) also known as posterior pituitary, stores and releases two hormones called oxytocin and vasopressin, which are actually synthesised by the hypothalamus and are transported axonally to neurohypophysis.

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